KERALA POLYTECHNIC · REVISION 2026

PRACTICUM | SEMESTER I/II | 4.5 CREDITS | 90 CONTACT HOURS

Applied Physics

Course 2002A — Mechanics · Properties of Matter · Optics · Electromagnetism

SUBJECT CODE	TEACHING SCHEME	CIE / ESE	SELF-LEARNING
2002A	**3:0:3:0 (L:T:P:R)**	**40 / 60**	**90 hours**

📄 Official Source Declaration

This handbook is derived from the Kerala Polytechnic Revision 2026 syllabus for **Course 2002A — Applied Physics**.

> **Source:** *2002A.pdf* — Semester I/II Practicum syllabus, 90 contact hours, 4.5 credits.
> **Status:** All modules, topics, subtopics, learning outcomes, self-learning activities, and assessment patterns have been faithfully extracted and elaborated for teaching purposes.

> ⚠️ **Verified Inconsistency:** The PDF lists "*3 hour per experiment*" in multiple places and also states "*A minimum of 8 to 10 experiments shall be conducted*" while listing 16+ experiments across modules. This handbook treats all listed experiments as prescribed, with the understanding that institutions may select a subset based on available equipment and time.

📊 Course Dashboard

🎯 Programme

Diploma in Engineering (all branches) · Semester I/II

📚 Course Type

Practicum · 3:0:3:0 (L:T:P:R) · 4.5 Credits

⏱️ Contact Hours

90 hours total · 3 hours per experiment · 16+ experiments

📝 Assessment

CIE 40 · ESE 60 · Series tests · Practical tests

Teaching & Assessment Scheme

Component	Details	Marks
Attendance	Minimum 75% required	5
CA1 (Practical test — first half)	End of semester (by 4th week)	10
CA2 (Practical test — second half)	By 7th week	10
CA3 (Series Exam — 2 modules)	By 11th week	10
CA4 (Series Exam — 2 modules)	By 14th week	10
CA5 (Average of self-learning activities)	Continuous	5
CIE Total		**40**
ESE (Written Exam — Theory)	2.5 hours · End of semester	**60**

💡 **Did you know?** The series examination pattern is: Part A (1 mark each), Part B (3 marks each), Part C (7 marks each) — covering all modules with choice.

🧭 How to Use This Handbook

1
Understand
Read each module's concepts, definitions, and principles. Use the Malayalam support notes for clarity.

2
Visualise
Study the labelled diagrams, formula cards, and worked examples. Use the interactive calculators to explore relationships.

3
Apply
Solve the numerical problems, attempt the self-learning activities, and review the module-wise Q&A.

4
Revise
Use the quick-revision cards, common-mistakes list, glossary, and practice model paper before exams.

🎯 Course Outcomes

By the end of this course, students will be able to:

Course Outcomes with Bloom's Taxonomy Level

Code	Course Outcome	Bloom's Level
CO1	Apply principles of mechanics (Newton's laws, work, energy, and power) to analyze motion-related problems and real-life engineering applications.	Apply
CO2	Explain and apply concepts of properties of matter, fluid dynamics, and thermodynamics (elasticity, viscosity, Bernoulli's principle, and heat transfer) to solve numerical problems and interpret practical phenomena.	Apply
CO3	Analyze rotational motion and wave phenomena (SHM, wave characteristics, and related relations) to solve engineering problems and interpret physical systems.	Apply
CO4	Apply principles of optics and electromagnetism (lens systems, electricity, magnetism, and electromagnetic induction) to solve numerical problems and explain working of devices.	Apply

CO–PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	—	2	2	2	—	—	—	—
CO2	3	2	2	—	2	2	2	—	—	—	—
CO3	3	2	2	—	2	2	2	—	—	—	—
Legends: 1 – Low, 2 – Medium, 3 – High, — No Correlation											

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO4	3	2	2	—	2	2	2	—	—	—	—
Legends: 1 – Low, 2 – Medium, 3 – High, — No Correlation											

🗺️ Curriculum Coverage Map

Every module, topic, and subtopic from the syllabus — mapped to hours, COs, and handbook anchors.

Module–Topic–Hours–CO Mapping

Module	Topics / Subtopics	Hours	CO	Anchor
I	Units, Scalars & Vectors, Laws of Motion, Work Energy & Power, Vernier Calipers, Screw Gauge, Spherometer, Parallelogram Law	~22	CO1	Module I
II	Elasticity, Fluid Dynamics, Heat & Temperature, Mercury Thermometer, Hooke's Law, Stoke's Law, U-tube Apparatus	~22	CO2	Module II
III	Circular Motion, Rotational Motion, SHM, Wave Motion, Simple Pendulum, Moment Bar, Spring Constant, Resonance Column, Flywheel	~23	CO3	Module III
IV	Reflection, Refraction, Lenses, Electricity, Magnetism & Electromagnetic Induction, Snell's Law, Convex Lens, Ohm's Law	~23	CO4	Module IV

🔗 How Modules Connect

⚙️ **Module I**
Mechanics & Measurement — Foundation of motion, forces, energy, and precision measurement tools.

🧪 **Module II**
Properties of Matter — Elasticity, fluid behaviour, heat transfer, and viscosity.

🌀 **Module III**
Rotational & Wave Motion — Circular motion, moment of inertia, SHM, and wave propagation.

💡 **Module IV**
Optics & Electromagnetism — Light, lenses, electric circuits, and electromagnetic induction.

MODULE I

Mechanics & Measurement

Units · Scalars & Vectors · Laws of Motion · Work, Energy & Power · Measurement Instruments

⏱️ ~22 hours 🎯 CO1 📊 Bloom: Understand / Apply

▸Units — Fundamental & Derived Quantities

Physical quantities are measurable properties of matter or phenomena. They are classified into two categories:

- **Fundamental quantities** — independent, base quantities (Length, Mass, Time, Temperature, Electric Current, Luminous Intensity, Amount of Substance).
- **Derived quantities** — obtained from fundamental quantities by multiplication or division (e.g., Velocity = Length/Time, Force = Mass × Acceleration).

Fundamental Quantities and Their SI Units

Fundamental Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric Current	ampere	A
Luminous Intensity	candela	cd
Amount of Substance	mole	mol

The **SI system** (International System of Units) is used worldwide for scientific and engineering measurements. It ensures consistency, reproducibility, and ease of communication.

> **Derived quantity example:** Force = Mass × Acceleration → Unit: $kg \cdot m/s^2$ = **newton (N)**

> **Malayalam support:** □□□□□□□□ □□□□□□ (Fundamental quantities) □□□□□□□□□□□□ □□□□□□□□□. □□□: □□□□, □□□□□□, □□□□. □□□□□ □□□□□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□ □□□□□□ (Derived quantities). □□□: □□□□□□□ (Velocity) = □□□□ / □□□□.

▸Scalars and Vectors — Triangle & Parallelogram Laws

Scalar quantities have only magnitude (e.g., mass, time, temperature, speed). **Vector quantities** have both magnitude and direction (e.g., velocity, force, displacement, acceleration).

Scalars vs. Vectors

Scalar	Vector
Mass, Time, Temperature, Speed, Energy, Work	Velocity, Force, Displacement, Acceleration, Momentum

Triangle Law of Vector Addition
If two vectors are represented by two sides of a triangle taken in order, then the resultant is given by the third side taken in the opposite order.

> $$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

where θ is the angle between the two vectors.
Parallelogram Law of Vector Addition
If two vectors are represented by adjacent sides of a parallelogram, the resultant is given by the diagonal of the parallelogram starting from the same point.

> $$R = \sqrt{A^2 + B^2 + 2AB \cos \theta} \quad \text{(same formula as triangle law)}$$

Resolution of vectors: A vector can be resolved into two perpendicular components: $V_x = V \cos \theta$ and $V_y = V \sin \theta$.

WORKED EXAMPLE
Given: Two forces of 3 N and 4 N act at an angle of 60°.
Required: Magnitude of the resultant.
Calculation: $R = \sqrt{3^2 + 4^2 + 2\times3\times4\times\cos60°} = \sqrt{9 + 16 + 12} = \sqrt{37} \approx 6.08$ N.
Answer: 6.08 N.

Malayalam support: □□□□□□□□□ (Vectors) □□ □□□□□□ □□□□□ □□□□□. □□□: □□□ (Force), □□□□□□□□ (Velocity). □□□□□□□□□ (Scalars) □□ □□□□ □□□□□□. □□□: □□□□□□, □□□□.

🧮 Vector Addition Calculator

Find the resultant of two vectors.

Vector A (magnitude)	Vector B (magnitude)	Angle θ (degrees)
5	3	60

Resultant: —

▸Newton's Laws of Motion — Momentum — Conservation

Newton's First Law (Law of Inertia)
An object remains at rest or in uniform motion unless acted upon by an external unbalanced force.
Newton's Second Law
The rate of change of momentum of an object is directly proportional to the applied force and takes place in the direction of the force.

> $$F = ma \quad \text{(where } F = \text{force, } m = \text{mass, } a = \text{acceleration)}$$

Momentum (p) is the product of mass and velocity: `p = mv`. Its SI unit is kg·m/s.
Newton's Third Law
For every action, there is an equal and opposite reaction.
Law of Conservation of Momentum
In a closed system with no external forces, the total momentum before and after an interaction remains constant.

> $$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Application: Recoil of a Gun
When a bullet is fired, the gun recoils backward. By conservation of momentum:

> $$m_{bullet} \times v_{bullet} = m_{gun} \times v_{gun}$$

The recoil velocity is opposite in direction to the bullet's motion.

WORKED EXAMPLE — RECOIL
Given: A 10 g bullet leaves a 2 kg gun at 400 m/s.
Required: Recoil velocity of the gun.
Calculation: $0.01 \times 400 = 2 \times v_{gun} \rightarrow v_{gun} = 4/2 = 2$ m/s.
Answer: 2 m/s (opposite direction to bullet).

> **Malayalam support:** □□□□□□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□□□, □□□□ (F) = □□□□□□□ (m) × □□□□□□□ (a). □□□□□□□ (Momentum) = □□□□□□□ × □□□□□□□□. □□□□□ □□□□□□□□ □□□□□□ (Conservation of momentum) □□□□□□□□, □□□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□□□.

▸Work, Energy, and Power — Kinetic & Potential Energy

Work (W)
Work is done when a force causes displacement. $W = F \times s \times \cos\theta$, where θ is the angle between force and displacement. Unit: **joule (J)**.

Energy (E)
Energy is the capacity to do work. Types: **Kinetic Energy** (energy of motion) and **Potential Energy** (stored energy).

> Kinetic Energy: $KE = \frac{1}{2} mv^2$ (m = mass, v = velocity)

> Potential Energy (gravitational): $PE = mgh$ (h = height)

Law of Conservation of Energy
Energy cannot be created or destroyed; it can only be transformed from one form to another. In a freely falling body, PE decreases while KE increases, but total energy remains constant.

Power (P)
Power is the rate of doing work. $P = W / t$. Unit: **watt (W)** = 1 J/s.

WORKED EXAMPLE — KINETIC ENERGY
Given: A 10 kg object moves at 5 m/s.
Required: Kinetic energy.
Calculation: $KE = \frac{1}{2} \times 10 \times 5^2 = 5 \times 25 = 125$ J.
Answer: 125 J.

WORKED EXAMPLE — POTENTIAL ENERGY
Given: A 5 kg mass is raised to a height of 3 m. ($g = 9.8$ m/s^2)
Required: Potential energy.
Calculation: $PE = 5 \times 9.8 \times 3 = 147$ J.
Answer: 147 J.

Malayalam support: □□□□□□□□□ (Work) = □□□ × □□□□□□□□□□□□. □□□□□□□ (Energy) □□□□□□ □□□□□□□□□: □□□□□□□□□□□ (KE) = ½ mv², □□□□□□□□□□□□□□ (PE) = mgh. □□□ (Power) = □□□□□□□□□□ / □□□□□.

▸Vernier Calipers · Screw Gauge · Spherometer · Parallelogram Law

Vernier Calipers

Used to measure internal and external dimensions with accuracy up to 0.01 cm (0.1 mm). It has a main scale and a vernier scale.

- **Least count (LC)** = 1 MSD − 1 VSD (typically 0.01 cm).
- Formula: `Reading = MSR + (VSR × LC)` .

Screw Gauge

Used to measure small diameters (wire, rod) and thickness with accuracy up to 0.001 cm (0.01 mm). It uses a screw with a fine pitch.

- **Least count** = Pitch / Number of divisions on circular scale.
- Formula: `Reading = MSR + (CSR × LC) − zero error` .

Spherometer

Measures the radius of curvature of spherical surfaces. It has a tripod with three fixed legs and a central screw with a circular scale.

- Formula: $R = (l^2 / 6h) + (h/2)$, where l = distance between legs, h = sagitta.

Parallelogram Law of Forces — Experiment

To verify the parallelogram law and find the mass of a given body. Two forces are applied at an angle, and the resultant is balanced by a third force. The resultant is measured and compared with the theoretical value.

> 🔧 **Lab Tip:** Always take multiple readings to minimize errors. Check for zero error before starting any measurement.

> **Malayalam support:** □□□□□□□□ □□□□□□□□□□□□, □□□□□□□ □□□□, □□□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□. □□□ □□□□□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□ □□□□ (Least count) □□□□□□□□□□□□□□□ □□□□□□□□□□□□.

> 📌 **Module I Summary:** Mechanics forms the foundation of physics. Key takeaways: SI units, vector addition, Newton's laws, conservation of momentum, work-energy-power relationships, and precision measurement techniques. Practise numerical problems on these topics regularly.

MODULE II

Properties of Matter

Elasticity · Fluid Dynamics · Heat & Temperature · Practical Experiments

⏱️ ~22 hours 🎯 CO2 📊 Bloom: Understand / Apply

▸Elasticity — Stress, Strain, Hooke's Law, Moduli

Elasticity is the property of a body to regain its original shape and size after the removal of deforming forces.

- **Stress (σ)** = Force / Area — the internal restoring force per unit area. Unit: N/m^2 or **pascal (Pa)**.
- **Strain (ε)** = Change in dimension / Original dimension — dimensionless.

Hooke's Law
Within the elastic limit, stress is directly proportional to strain.

> Stress ∝ Strain → $\sigma = E \times \varepsilon$ (E = modulus of elasticity)

Types of Moduli

- **Young's Modulus (Y)** — for longitudinal stress/strain: $Y = (F/A) / (\Delta L/L)$.
- **Bulk Modulus (K)** — for volume stress/strain: $K = (\Delta P) / (\Delta V/V)$.
- **Shear Modulus (η or G)** — for tangential stress/strain: $\eta = (F/A) / (\Delta x/h)$.

WORKED EXAMPLE — YOUNG'S MODULUS
Given: A wire of length 2 m, cross-section area 1×10^{-6} m^2, stretches by 0.5 mm when a 100 N load is applied.
Required: Young's modulus.
Calculation: Stress = $100/1\times10^{-6} = 10^8$ Pa. Strain = $0.5\times10^{-3}/2 = 2.5\times10^{-4}$. $Y = 10^8 / 2.5\times10^{-4} = 4\times10^{11}$ Pa.
Answer: 4×10^{11} Pa.

> **Malayalam support:** [illegible] (Stress) = [illegible] / [illegible]. [illegible] (Strain) = [illegible] / [illegible] [illegible]. [illegible] [illegible]: [illegible] ∝ [illegible]. [illegible] [illegible] (Young's modulus) [illegible] [illegible] [illegible] [illegible] [illegible].

▸Fluid Dynamics — Streamline/Turbulent Flow · Continuity · Bernoulli · Viscosity

Streamline vs. Turbulent Flow

- **Streamline flow** — smooth, ordered flow where each particle follows a path that does not intersect with others.
- **Turbulent flow** — irregular, chaotic flow with eddies and vortices.

Equation of Continuity

For an incompressible fluid in steady flow, the product of cross-sectional area and velocity remains constant along a stream tube.

> $$A_1 v_1 = A_2 v_2 \quad (A = \text{area}, v = \text{velocity})$$

Bernoulli's Principle

For an ideal fluid in steady flow, the sum of pressure energy, kinetic energy per unit volume, and potential energy per unit volume is constant along a streamline.

> $$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Application: Atomizer — high-speed air over a tube reduces pressure, causing liquid to be drawn up and sprayed.

Viscosity and Stoke's Law

Viscosity is the internal friction between layers of a fluid. **Stoke's Law** gives the viscous force on a spherical body moving through a fluid.

> $$F = 6\pi\eta r v \quad (\eta = \text{coefficient of viscosity}, r = \text{radius}, v = \text{velocity})$$

Terminal velocity is reached when the viscous force balances the net weight of the body.

> **Malayalam support:** [illegible]: [illegible]. [illegible] (Viscosity) [illegible]. [illegible]: F = 6πηrv.

▸Heat vs. Temperature · Scales · Heat Transfer

Heat is a form of energy transferred due to a temperature difference.
Temperature is a measure of the average kinetic energy of particles.

Temperature Scales

- **Celsius (°C)** — water freezes at 0°C, boils at 100°C.
- **Fahrenheit (°F)** — water freezes at 32°F, boils at 212°F.
- **Kelvin (K)** — SI unit, 0 K = absolute zero (−273.15°C).

$$T(K) = T(°C) + 273.15$$

$$T(°F) = (9/5) \times T(°C) + 32$$

Modes of Heat Transfer

- **Conduction** — transfer through molecular collisions (solids).
- **Convection** — transfer by bulk movement of fluid (liquids/gases).
- **Radiation** — transfer through electromagnetic waves (no medium needed).

WORKED EXAMPLE — TEMPERATURE CONVERSION
Given: Convert 37°C to Fahrenheit and Kelvin.
Calculation: °F = (9/5)×37 + 32 = 66.6 + 32 = 98.6°F. K = 37 + 273.15 = 310.15 K.
Answer: 98.6°F, 310.15 K.

Malayalam support: □□□□ (Heat) □□□□□□ □□□□□□□□□□□□□ □□□ □□□□□□□. □□□□□□ (Temperature) □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□. □□□□ □□□□□□□□□□□□□ □□□□□□ □□□□□□: □□□□□ (Conduction), □□□□□□ (Convection), □□□□□□□ (Radiation).

▸Experiments — Mercury Thermometer · Hooke's Law · Stoke's Law · U-tube

Mercury Thermometer

Used to measure room temperature and hot bath temperature. Mercury expands uniformly with temperature, and the reading is taken from the calibrated scale.

Hooke's Law Experiment

Determine the spring constant by applying loads and measuring extension. Plot a graph of load vs. extension — the slope gives the spring constant (k).

> $$F = kx \quad (k = \text{spring constant},\ x = \text{extension})$$

Stoke's Law Experiment

Measure the coefficient of viscosity of a liquid by observing the terminal velocity of a small spherical body falling through the liquid.

> $$\eta = (2/9) \times (r^2 g(\rho - \sigma)) / v_t$$

where ρ = density of sphere, σ = density of liquid, v_t = terminal velocity.

U-tube Apparatus

Used to determine the relative density (specific gravity) of a liquid by comparing the heights of two liquid columns in a U-tube.

> 🔬 **Lab Tip:** Ensure the apparatus is clean and free from air bubbles. Take multiple readings and average them to minimise errors.

> **Malayalam support:** [illegible], [illegible] (load) vs. [illegible] (extension) [illegible] (k) [illegible]. [illegible].

> 📌 **Module II Summary:** Properties of matter are fundamental to engineering materials and fluid systems. Key concepts: Hooke's law, Young's modulus, Bernoulli's principle, viscosity, heat transfer modes. Practise numerical problems on stress/strain, temperature conversion, and viscosity.

MODULE III

Rotational & Wave Motion

Circular Motion · Rotational Motion · SHM · Wave Motion · Practical Experiments

⏱️ ~23 hours 🎯 CO3 📊 Bloom: Understand / Apply

▸Circular Motion — Angular Displacement, Velocity, Acceleration

In **circular motion**, an object moves along a circular path. The key parameters are:

- **Angular displacement (θ)** — angle swept by the radius vector (radians).
- **Angular velocity (ω)** — rate of change of angular displacement: $\omega = \Delta\theta/\Delta t$ (rad/s).
- **Angular acceleration (α)** — rate of change of angular velocity: $\alpha = \Delta\omega/\Delta t$ (rad/s²).

Relation between Linear and Angular Quantities

$$v = r\omega \quad \text{(linear velocity = radius × angular velocity)}$$

$$a = r\alpha \quad \text{(tangential acceleration = radius × angular acceleration)}$$

Centripetal Acceleration and Centripetal Force

In uniform circular motion, the acceleration is directed towards the centre.

$$a_c = v^2/r = r\omega^2$$

$$F_c = m\,v^2/r = m\,r\,\omega^2$$

Malayalam support: □□□□□□□□□ □□□□□□□□, □□□□□ □□□□□□□ (ω) = Δθ/Δt. □□□□□ □□□□□□□ v = rω. □□□□□□□□□□□□□ (Centripetal force) $F_c = m v^2/r$.

▸Rotational Motion — Moment of Inertia · Radius of Gyration · Angular Momentum

Rigid body — a body where the relative positions of all particles remain fixed.

Moment of Inertia (I)

It is the rotational analogue of mass. For a system of particles: $I = \Sigma m_i r_i^2$.

Radius of Gyration (k)

The distance from the axis at which the total mass can be considered to be concentrated: $I = Mk^2$.

Parallel and Perpendicular Axis Theorems

- **Parallel axis theorem:** $I = I_{cm} + Md^2$ (d = distance between axes).
- **Perpendicular axis theorem:** $I_z = I_x + I_y$ (for a plane lamina).

Moment of Inertia for Common Bodies

- Rod (centre): $I = (1/12)ML^2$
- Disc (axis through centre perpendicular to plane): $I = (1/2)MR^2$
- Ring (axis through centre perpendicular to plane): $I = MR^2$
- Solid sphere: $I = (2/5)MR^2$
- Hollow sphere: $I = (2/3)MR^2$

Angular Momentum (L) and Torque (τ)

> $L = I\omega$ (angular momentum = moment of inertia × angular velocity)

> $\tau = I\alpha$ (torque = moment of inertia × angular acceleration)

Conservation of Angular Momentum: In the absence of external torque, total angular momentum is conserved. (Example: an ice skater spinning faster by pulling arms in.)

> **Malayalam support:** □□□□□□□□□□ (Moment of inertia) $I = \Sigma m_i r_i^2$. □□□□□ □□□□□ (Angular momentum) $L = I\omega$. □□□□□ □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□ □□□□□□□□□□□□□□□□.

▸Simple Harmonic Motion (SHM) — Periodic Motion · Time Period · Frequency

Periodic motion repeats itself after a fixed interval of time. **Simple Harmonic Motion (SHM)** is a type of periodic motion where the restoring force is directly proportional to displacement and directed towards the mean position.

> $F = -kx$ (k = force constant, x = displacement)

For SHM: $x = A \sin(\omega t + \varphi)$, where A = amplitude, ω = angular frequency.

- **Time period (T)** = $2\pi/\omega = 2\pi\sqrt{(m/k)}$
- **Frequency (f)** = $1/T = \omega/2\pi$

Examples: spring-mass system, simple pendulum, vibrations in machines.

> **Malayalam support:** SHM-□, □□□□□□□□□□□□□ (restoring force) □□□□□□□□□□□□□□□□ (displacement) □□□□□□□□□□□□. $T = 2\pi\sqrt{(m/k)}$. □□□: □□□□□□□□□□□□ □□□□, □□□□□□□□□-□□□□ □□□□□□□□.

▸Wave Motion — Transverse/Longitudinal · Frequency, Velocity, Wavelength

Types of Waves

- **Transverse waves** — particles oscillate perpendicular to direction of propagation (e.g., waves on a string, light waves).
- **Longitudinal waves** — particles oscillate parallel to direction of propagation (e.g., sound waves).

Wave Characteristics

- **Frequency (f)** — number of oscillations per second (Hz).
- **Wavelength (λ)** — distance between two consecutive points in phase.
- **Wave velocity (v)** — speed of propagation.

> ```
> v = fλ   (wave velocity = frequency × wavelength)
> ```

WORKED EXAMPLE — WAVE VELOCITY
Given: A wave of frequency 500 Hz has a wavelength of 0.7 m.
Required: Wave velocity.
Calculation: v = 500 × 0.7 = 350 m/s.
Answer: 350 m/s.

> **Malayalam support:** □□□□ □□□□□□□ (v) = □□□□□□□ (f) × □□□□□□□□□□□ (λ). □□□□□□□□□ □□□□□□□□ (Transverse waves) — □□□□□□ □□□□□□ □□□□□□□□□ □□□□□□□ □□□□□□□□□□□. □□□□□□□□□ □□□□□□□□ (Longitudinal waves) — □□□□□□ □□□□□□ □□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□.

▸Experiments — Simple Pendulum · Moment Bar · Spring Constant · Resonance Column · Flywheel

Simple Pendulum

Determine acceleration due to gravity (g) by measuring the time period (T) for different lengths (L).

> $$T = 2\pi\sqrt{(L/g)} \rightarrow g = 4\pi^2 L/T^2$$

Moment Bar

Determine the mass of a given body using the principle of moments. `Clockwise moment = Anticlockwise moment`.

Spring Constant (Oscillation Method)

Determine the spring constant (k) by measuring the time period of oscillations: $T = 2\pi\sqrt{(m/k)}$.

Resonance Column

Determine the velocity of sound in air at room temperature by finding the resonant lengths of the air column for a tuning fork of known frequency.

> $$v = 4f(L_1 + 0.3d) \quad (d = \text{diameter of tube})$$

Flywheel

Find the moment of inertia of a flywheel by observing the motion of a falling weight attached to a string wound around the axle.

> 🔧 **Lab Tip:** In the resonance column experiment, ensure the water level is adjusted slowly to find the exact resonant position. Take multiple readings for accuracy.

> **Malayalam support:** [illegible], $T = 2\pi\sqrt{(L/g)}$ [illegible] g [illegible]. [illegible].

> 📌 **Module III Summary:** Rotational and wave motion are crucial for understanding machinery, vibrations, and sound. Key concepts: angular velocity/acceleration, moment of inertia, SHM, wave equations. Practise numerical problems on wave velocity, SHM time period, and moment of inertia.

MODULE IV

Optics & Electromagnetism

Reflection · Refraction · Lenses · Electricity · Magnetism · Electromagnetic Induction

⏱️ ~23 hours 🎯 CO4 📊 Bloom: Understand / Apply

▸Reflection — Laws of Reflection

Reflection is the bouncing back of light when it strikes a surface.

Laws of Reflection

1. The incident ray, reflected ray, and normal all lie in the same plane.
2. The angle of incidence equals the angle of reflection (∠i = ∠r).

> **Malayalam support:** [illegible]: (1) [illegible], [illegible], [illegible]. (2) [illegible] (∠i = ∠r).

▸Refraction — Snell's Law · Refractive Index

Refraction is the bending of light when it passes from one medium to another.

Snell's Law

$$n_1 \sin i = n_2 \sin r$$

where n_1, n_2 are refractive indices of the two media, i = angle of incidence, r = angle of refraction.

Refractive Index (n)

The ratio of speed of light in vacuum to speed of light in the medium: `n = c/v` . Absolute refractive index of air ≈ 1.0003, glass ≈ 1.5, water ≈ 1.33.

WORKED EXAMPLE — SNELL'S LAW

Given: Light enters glass (n_2 = 1.5) from air (n_1 = 1.0) at an angle of incidence 30°.

Required: Angle of refraction.

Calculation: $1.0 \times \sin 30° = 1.5 \times \sin r \rightarrow \sin r = 0.5/1.5 = 0.333 \rightarrow r = \sin^{-1}(0.333) \approx 19.5°$.

Answer: 19.5°.

Malayalam support: □□□□□□□□□□□□ □□□□□□: $n_1 \sin i = n_2 \sin r$. □□□□□□□□□□□□□□ (Refractive index) n = c/v. □□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□, □□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□.

▶Lenses — Convex & Concave · Focal Length · Lens Formula · Power

- **Convex lens** — converges parallel rays to a focal point. (Converging lens).
- **Concave lens** — diverges parallel rays. (Diverging lens).

Focal Length (f)

Distance from the optical centre to the focal point. For a convex lens, f is positive; for a concave lens, f is negative (sign convention).

Lens Formula

$$1/f = 1/v - 1/u$$

where u = object distance (negative), v = image distance (positive for real image).

Power of a Lens (P)

Power is the reciprocal of focal length in metres: `P = 1/f`. Unit: **dioptre (D)**.

WORKED EXAMPLE — LENS FORMULA

Given: An object is placed 20 cm from a convex lens of focal length 10 cm.
Required: Image distance and power.
Calculation: $1/f = 1/v - 1/u \rightarrow 1/10 = 1/v - 1/(-20) \rightarrow 1/v = 1/10 - 1/20 = 1/20 \rightarrow v = 20$ cm. Power = 1/0.1 = 10 D.
Answer: v = 20 cm, P = 10 D.

Malayalam support: □□□□□□□□□□ □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ (converging). □□□□□□□□ □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□ (diverging). □□□□□ □□□□□□□□□□□□: 1/f = 1/v − 1/u. □□□ P = 1/f (dioptre).

▸Electricity — Charge · Coulomb's Law · Electric Field · Ohm's Law · Resistance

Electric Charge and Coulomb's Law
Charge is a fundamental property of matter. Like charges repel, unlike charges attract. Coulomb's law gives the force between two charges.

> $F = k \times (q_1 q_2) / r^2 \quad (k = 9\times10^9\ N\cdot m^2/C^2)$

Electric Field and Potential
Electric field (E) = force per unit charge: $E = F/q$. Unit: N/C or V/m.
Electric potential (V) = work done per unit charge to bring a charge from infinity. Unit: volt (V).
Electric Current
Current is the rate of flow of charge: $I = Q/t$. Unit: **ampere (A)**.

- **Direct current (DC)** — flows in one direction (e.g., battery).
- **Alternating current (AC)** — changes direction periodically (e.g., mains supply).

Ohm's Law

> $V = IR \quad (V = \text{voltage}, I = \text{current}, R = \text{resistance})$

Resistance is the opposition to current flow. Unit: **ohm (Ω)**.

WORKED EXAMPLE — OHM'S LAW
Given: A 12 V battery supplies a current of 2 A.
Required: Resistance of the circuit.
Calculation: R = V/I = 12/2 = 6 Ω.
Answer: 6 Ω.

🧮 **Ohm's Law Calculator**
Calculate Voltage, Current, or Resistance.

Voltage (V)	Current (A)	Resistance (Ω)
12	2	R

Enter any two values to calculate the third.

Malayalam support: [illegible] [illegible]: $F = kq_1q_2/r^2$. [illegible] [illegible] $I = Q/t$. [illegible] [illegible]: $V = IR$. [illegible] [illegible] (DC) [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] (AC) [illegible] [illegible] [illegible] [illegible].

▸Magnetism · Magnetic Flux · Lorentz Force · Faraday's Laws

Magnetic Field and Flux

Magnetic field is the region around a magnet where force is experienced. **Magnetic flux (Φ)** is the number of field lines passing through a surface. Unit: **weber (Wb)**.

Lorentz Force

The force on a charged particle moving in a magnetic field:

```
F = q(v × B)   (q = charge, v = velocity, B = magnetic field)
```

Electromagnetic Induction — Faraday's Laws

When magnetic flux through a coil changes, an EMF is induced. Faraday's laws state:

- The induced EMF is proportional to the rate of change of flux.
- The magnitude of induced EMF: $\varepsilon = -\, d\Phi/dt$ (Lenz's law gives the direction).

Applications: Generators, transformers, induction motors.

Malayalam support: [illegible] [illegible]: [illegible] [illegible] [illegible] [illegible] EMF-[illegible] [illegible]. $\varepsilon = -\, d\Phi/dt$. [illegible] [illegible], [illegible] [illegible] [illegible] [illegible].

▸Experiments — Snell's Law · Convex Lens (u-v method) · Ohm's Law

Snell's Law Experiment
Verify Snell's law using a glass slab or glass prism. Trace the path of light, measure the angles of incidence and refraction, and verify that $n_1 \sin i = n_2 \sin r$.

Convex Lens — u-v Method
Determine the focal length and power of a convex lens by the u-v method (object distance/image distance method). Plot a graph of 1/v vs. 1/u to find the focal length from the intercept.

Ohm's Law Experiment
Verify Ohm's law by plotting a graph between current (I) and potential difference (V). The graph should be a straight line passing through the origin, with slope = 1/R.

> 🔬 **Lab Tip:** In the Ohm's law experiment, change the resistance values and take multiple readings to plot a reliable V-I graph. Ensure connections are tight and the voltmeter/ammeter are correctly polarised.

> **Malayalam support:** [illegible], [illegible]/ [illegible]. [illegible], V vs I [illegible].

> 📌 **Module IV Summary:** Optics and electromagnetism are fundamental to modern technology. Key concepts: Snell's law, lens formula, Ohm's law, Faraday's laws. Practise numerical problems on lens formula, power, and Ohm's law. Understand the working of generators and transformers.

📐 Formula Bank

Quick reference for all important formulas from the course.

$F = ma$
Newton's second law (Force = mass × acceleration)

$p = mv$
Momentum = mass × velocity

$KE = \frac{1}{2}mv^2$
Kinetic energy

$PE = mgh$

Gravitational potential energy

$P = W/t$

Power = work / time

$\sigma = F/A$

Stress = force / area

$\varepsilon = \Delta L/L$

Strain = change in length / original length

$Y = \sigma/\varepsilon$

Young's modulus = stress / strain

$A_1v_1 = A_2v_2$

Equation of continuity

$P + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$

Bernoulli's principle

$F = 6\pi\eta rv$

Stoke's law (viscous force)

$T = 2\pi\sqrt{(L/g)}$

Simple pendulum time period

$T = 2\pi\sqrt{(m/k)}$

Spring-mass system time period

$v = f\lambda$

Wave velocity = frequency × wavelength

$n_1 \sin i = n_2 \sin r$

Snell's law of refraction

$1/f = 1/v - 1/u$

Lens formula

$P = 1/f$

Power of lens (dioptre)

$V = IR$

Ohm's law

$\varepsilon = -d\Phi/dt$

Faraday's law of induction

$F = q(v \times B)$

Lorentz force

🖼️ Diagram Library for Rapid Revision

Key diagrams from each module — click a topic to jump to its full explanation.

📐 Vector Addition

Triangle & Parallelogram laws

Jump to Module I →

⚙️ Hooke's Law

Stress–strain relationship

Jump to Module II →

🌀 Wave Motion

Transverse vs. Longitudinal

Jump to Module III →

💡 Refraction

Snell's law — bending of light

Jump to Module IV →

📝 Self-Learning Activities

Official activities from the syllabus — 90 hours across 16 weeks.

WEEK 1–4

Module I Activities

- Classify quantities as fundamental/derived with SI units.
- Solve numerical problems on vector addition and resolution.
- Demonstrate Newton's first law with real-life examples.
- Calculate energy consumption of household appliances.

WEEK 5–8

Module II Activities

- Hooke's law at home using rubber band and coins.
- Identify 5 real-life applications of elasticity.
- Water flow from bottle — verify continuity equation.
- Compare viscosity of different liquids (water, oil, honey).
- Temperature conversion and heat transfer examples.

WEEK 9–12

Module III Activities

- Identify real-life circular motion examples.
- Watch YouTube simulations on SHM — note amplitude/frequency.
- Create comparison chart of moment of inertia for different bodies.
- Solve numerical problems on wave velocity, wavelength, frequency.
- Use PhET simulations on waves on a string.

WEEK 13–16

Module IV Activities

- Observe practical examples of refraction.
- List devices using convex and concave lenses.
- Solve 10 numerical problems on lens formula and power.
- Study why AC is preferred over DC for power transmission.
- Analyse household electrical devices — type of current, voltage, resistance.

📊 Assessment Methodology

Practicum Assessment — CIE and ESE

Mode	Description	Duration	Marks (Converted)
Attendance	Minimum 75% required	—	5
CA1	Practical test (first half of experiments)	3 hours	10
CA2	Practical test (second half of experiments)	3 hours	10
CA3	Series Exam (2 modules)	2 hours	10
CA4	Series Exam (2 modules)	2 hours	10
CA5	Average of self-learning activities from each module	—	5
CIE Total			**40**
ESE	Written Exam (Theory)	2.5 hours	**60**

📝 Series Examination Pattern (Theory)

Series Test — 2 hours, 25 marks

Part	Type	Marks
Part A	Questions from any one module (1 mark each)	2
Part B	Questions from any one module (3 marks each)	3
Part C	Questions with choice from any one module (7 marks each)	3

📝 End Semester Examination Pattern (Theory)

ESE — 2.5 hours, 60 marks

Part	Description	No. of Questions	Marks

Part	Description	No. of Questions	Marks
Part A	2 questions from each module (1 mark each)	8	8
Part B	2 out of 3 questions from each module (3 marks each)	8	24
Part C	1 set questions with choice from each module (7 marks each)	4	28
Total		**20**	**60**

❓ Module-wise Questions with Answers

Practice questions covering all modules. Click a question to reveal its answer.

Module I — Mechanics & Measurement

Q: What are fundamental and derived quantities? Give examples.

A: Fundamental quantities are independent base quantities (Length, Mass, Time, Temperature, Electric Current, Luminous Intensity, Amount of Substance). Derived quantities are obtained from fundamental quantities (e.g., Velocity = Length/Time, Force = Mass × Acceleration).

Q: State and explain the triangle law of vector addition.

A: If two vectors are represented by two sides of a triangle taken in order, the resultant is given by the third side taken in the opposite order. Magnitude: $R = \sqrt{(A^2 + B^2 + 2AB \cos \theta)}$. Direction: $\tan \varphi = (B \sin \theta) / (A + B \cos \theta)$.

Q: State Newton's laws of motion. Explain conservation of momentum with a suitable example.

A: 1st Law: Inertia. 2nd Law: $F = ma$. 3rd Law: Action–Reaction. Conservation of momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. Example: Recoil of a gun — total momentum before firing = total momentum after firing.

Q: Define work, energy, and power. State their SI units.

A: Work (W) = $F \times s \times \cos \theta$ (J). Energy = capacity to do work (J). Power = rate of doing work (W = J/s).

Q: What is the least count of a vernier caliper and a screw gauge?

A: Vernier calipers: LC = 1 MSD − 1 VSD (typically 0.01 cm). Screw gauge: LC = Pitch / Number of divisions on circular scale (typically 0.001 cm).

Module II — Properties of Matter

Q: Define stress and strain. State Hooke's law.

A: Stress = Force / Area. Strain = Change in dimension / Original dimension. Hooke's law: Within elastic limit, stress ∝ strain. $\sigma = E \times \varepsilon$, where E is the modulus of elasticity.

Q: What is Bernoulli's principle? Give one application.

A: Bernoulli's principle: $P + \frac{1}{2}\rho v^2 + \rho g h$ = constant along a streamline. Application: Atomizer — high-speed air reduces pressure, drawing liquid up.

Q: What is viscosity? State Stoke's law.

A: Viscosity is the internal friction between layers of a fluid. Stoke's law: $F = 6\pi\eta r v$, where η = coefficient of viscosity, r = radius, v = velocity.

Q: Distinguish between heat and temperature.

A: Heat is a form of energy transferred due to temperature difference. Temperature is a measure of the average kinetic energy of particles.

Q: Explain the three modes of heat transfer.

A: Conduction (through solids by molecular collisions), Convection (through fluids by bulk movement), Radiation (through electromagnetic waves, no medium needed).

Module III — Rotational & Wave Motion

Q: Define angular displacement, angular velocity, and angular acceleration.

A: Angular displacement (θ) — angle swept (radians). Angular velocity (ω) = $\Delta\theta/\Delta t$ (rad/s). Angular acceleration (α) = $\Delta\omega/\Delta t$ (rad/s²).

Q: Derive the relation between linear and angular velocity.

A: $v = r\omega$. Linear velocity = radius × angular velocity.

Q: What is moment of inertia? State the parallel and perpendicular axis theorems.

A: Moment of inertia (I) = $\Sigma m_i r_i^2$. Parallel axis theorem: $I = I_{cm} + Md^2$. Perpendicular axis theorem: $I_z = I_x + I_y$ (for a plane lamina).

Q: Define simple harmonic motion. Write the expression for its time period.

A: SHM: restoring force ∝ displacement and directed towards mean position. $F = -kx$. Time period $T = 2\pi\sqrt{(m/k)}$ for a spring-mass system, $T = 2\pi\sqrt{(L/g)}$ for a simple pendulum.

Q: What is the relation between frequency, wave velocity, and wavelength?

A: $v = f\lambda$. Wave velocity = frequency × wavelength.

Module IV — Optics & Electromagnetism

Q: State Snell's law of refraction.

A: $n_1 \sin i = n_2 \sin r$, where n_1, n_2 are refractive indices, i = angle of incidence, r = angle of refraction.

Q: Write the lens formula and explain the sign convention.

A: $1/f = 1/v - 1/u$. Sign convention: u (object distance) is negative, v (image distance) is positive for real image, f is positive for convex lens.

Q: Define power of a lens. What is its unit?

A: Power $P = 1/f$, where f is in metres. Unit: dioptre (D).

Q: State Ohm's law and write the formula for resistance in a series circuit.

A: Ohm's law: $V = IR$. Series resistance: $R_s = R_1 + R_2 + R_3 + \ldots$

Q: State Faraday's laws of electromagnetic induction.

A: First law: An EMF is induced when magnetic flux changes. Second law: Induced EMF is proportional to the rate of change of flux. $\varepsilon = -d\Phi/dt$.

📝 Practice Model Paper

Based on the official ESE pattern — 60 marks, 2.5 hours.

> ⚠️ **Note:** This is a practice paper prepared by POLY PMNA for study purposes. It is **not** an official SITTTR model question paper.

Part A — (8 × 1 = 8 marks)

Answer all questions. Each question carries 1 mark.

1. What is the SI unit of force?
2. Define stress.

3. What is the relation between linear and angular velocity?
4. State Snell's law.
5. What is the power of a lens of focal length 0.5 m?
6. What is the SI unit of resistance?
7. Define momentum.
8. What is the time period of a simple pendulum of length 1 m? ($g = 9.8\ m/s^2$)

Part B — (8 × 3 = 24 marks)

Answer any 2 out of 3 questions from each module.

Module I

1. Explain the parallelogram law of vector addition. Find the resultant of two forces of 4 N and 3 N acting at 90°.
2. State the law of conservation of momentum. Explain with the example of recoil of a gun.
3. A 5 kg object is raised to a height of 4 m. Calculate its potential energy. ($g = 10\ m/s^2$)

Module II

1. Explain Hooke's law and define Young's modulus. A wire of length 3 m extends by 1 mm when a load of 200 N is applied. If the cross-section area is $2\times10^{-6}\ m^2$, find Young's modulus.
2. State Bernoulli's principle and explain its application in an atomizer.
3. Convert 100°C to Fahrenheit and Kelvin.

Module III

1. Define angular velocity and centripetal acceleration. Derive the relation $a_c = v^2/r$.
2. What is SHM? Derive the time period of a simple pendulum.
3. A wave of frequency 400 Hz travels at 320 m/s. Find its wavelength.

Module IV

1. An object is placed 15 cm from a convex lens of focal length 10 cm. Find the image distance and magnification.
2. State Ohm's law. A 12 V battery supplies a current of 3 A. Find the resistance.
3. State Faraday's laws of electromagnetic induction. Explain how a generator works.

Part C — (4 × 7 = 28 marks)

Answer any one set from each module. Each set has a choice.

Module I

Set 1: (a) Derive the lens formula. (b) A 10 g bullet is fired from a 2 kg gun at 300 m/s. Find the recoil velocity of the gun.

Set 2: (a) Explain Newton's laws of motion with examples. (b) A body of mass 2 kg is moving with a velocity of 5 m/s. Calculate its kinetic energy and the work done to bring it to rest.

Module II

Set 1: (a) Explain stress, strain, and Hooke's law. (b) A copper wire of length 2 m and radius 0.5 mm stretches by 0.2 mm under a load of 100 N. Find Young's modulus of copper.

Set 2: (a) Derive Bernoulli's equation and state its applications. (b) Explain the difference between streamline and turbulent flow with examples.

Module III

Set 1: (a) Derive the expression for the time period of a simple pendulum. (b) A body of mass 0.5 kg attached to a spring oscillates with a time period of 0.4 s. Find the spring constant.

Set 2: (a) Define moment of inertia. State the parallel axis theorem. (b) A wave has a frequency of 500 Hz and a wavelength of 0.6 m. Find its velocity and time period.

Module IV

Set 1: (a) Derive the lens formula $1/f = 1/v - 1/u$. (b) A convex lens forms a real image twice the size of the object. If the object distance is 15 cm, find the focal length.

Set 2: (a) State and explain Faraday's laws of electromagnetic induction. (b) A coil of 100 turns has a magnetic flux changing at 0.05 Wb/s. Find the induced EMF.

⚡ Quick Revision

Condensed key points for last-minute revision.

📌 Module I — Mechanics

- **SI units:** m, kg, s, A, K, cd, mol
- **Vector addition:** $R = \sqrt{A^2+B^2+2AB\cos\theta}$
- **Newton's laws:** Inertia, $F=ma$, Action–Reaction
- **Momentum:** $p=mv$, conservation: $m_1u_1+m_2u_2 = m_1v_1+m_2v_2$
- **Energy:** $KE=\frac{1}{2}mv^2$, $PE=mgh$, $Work=F\times s\times\cos\theta$
- **Power:** $P=W/t$

📌 Module II — Properties of Matter

- **Stress:** $\sigma=F/A$, **Strain:** $\varepsilon=\Delta L/L$
- **Hooke's law:** $\sigma = E\varepsilon$ (Y, K, η moduli)
- **Bernoulli:** $P+\frac{1}{2}\rho v^2+\rho gh = const$
- **Continuity:** $A_1v_1 = A_2v_2$
- **Viscosity:** $F=6\pi\eta rv$ (Stoke's law)
- **Temperature scales:** K = °C+273.15, °F = 9/5°C+32

📌 Module III — Rotational & Wave

- **Angular:** $\omega=\Delta\theta/\Delta t$, $\alpha=\Delta\omega/\Delta t$, $v=r\omega$
- **Centripetal:** $a_c=v^2/r=r\omega^2$, $F_c=mv^2/r$
- **Moment of inertia:** $I=\Sigma mr^2$
- **Angular momentum:** $L=I\omega$, $\tau=I\alpha$
- **SHM:** $F=-kx$, $T=2\pi\sqrt{m/k}$, $T=2\pi\sqrt{L/g}$
- **Wave:** $v=f\lambda$

📌 Module IV — Optics & Electromagnetism

- **Snell's law:** $n_1 \sin i = n_2 \sin r$
- **Lens formula:** $1/f=1/v-1/u$
- **Power:** $P=1/f$ (dioptre)
- **Ohm's law:** $V=IR$

- **Coulomb's law:** $F=kq_1q_2/r^2$
- **Faraday's law:** $\varepsilon=-d\Phi/dt$
- **Lorentz force:** $F=q(v\times B)$

⚠️ Common Mistakes to Avoid

In Mechanics

- Forgetting to use the correct sign convention for vectors.
- Confusing mass and weight (weight = mg).
- Not converting units to SI before calculations.

In Optics & Electricity

- Using the wrong sign convention in lens formula.
- Confusing series and parallel resistance formulas.
- Forgetting to convert cm to m in lens power calculations.

✅ Exam Checklist

- ☐ Know all SI units and their symbols.
- ☐ Memorise all formula cards from the formula bank.
- ☐ Practise numerical problems from each module.
- ☐ Understand the working of key experiments (pendulum, Ohm's law, lens).
- ☐ Review the CO–PO mapping for exam preparation.
- ☐ Use the glossary for quick term definitions.

📖 Glossary

Important terms and their meanings.

Key Terms in Applied Physics

Term	Meaning
Scalar	Quantity with only magnitude (e.g., mass, time).
Vector	Quantity with both magnitude and direction (e.g., force, velocity).
Momentum	Product of mass and velocity ($p = mv$).
Stress	Force per unit area ($\sigma = F/A$).
Strain	Change in dimension per original dimension.
Elasticity	Property of regaining original shape after removing deforming force.
Viscosity	Internal friction between layers of a fluid.
Angular Velocity	Rate of change of angular displacement ($\omega = \Delta\theta/\Delta t$).
Moment of Inertia	Rotational analogue of mass ($I = \Sigma mr^2$).
SHM	Simple Harmonic Motion — restoring force ∝ displacement.

Term	Meaning
Refraction	Bending of light when passing from one medium to another.
Refractive Index	Ratio of speed of light in vacuum to that in a medium (n = c/v).
Power (lens)	Reciprocal of focal length in metres (P = 1/f).
Resistance	Opposition to current flow (R = V/I).
Electromagnetic Induction	Production of EMF by changing magnetic flux.

📚 References

Textbooks

- NCERT — *Text Book of Physics for Class XI & XII (Part-I, Part-II)*
- Dr. H H Lal — *Applied Physics, Vol. I and Vol. II*, TTTI Publications, Tata McGraw Hill

Reference Books

- Halliday, Resnick and Walker — *Fundamentals of Physics*, Wiley India Pvt. Ltd

Web Resources

- — All modules
-

'; count++; } html += '

'; searchResults.innerHTML = html; // Click handler to open details searchResults.querySelectorAll('a[data-target-id]').forEach(link => { link.addEventListener('click', function(e) { const targetId = this.dataset.targetId; if (targetId) { const target = document.getElementById(targetId); if (target) { const details = target.closest('details'); if (details) details.open = true; setTimeout(() => { target.scrollIntoView({ behavior: 'smooth', block: 'start' }); }, 100); } } }); }); } if (searchBtn) { searchBtn.addEventListener('click', performSearch); } if (searchInput) { searchInput.addEventListener('keydown', function(e) { if (e.key === 'Enter') performSearch(); }); } // ---- REDUCED MOTION SAFETY ---- if (window.matchMedia('(prefers-reduced-motion:reduce)').matches) { document.querySelectorAll('.rotating, .slow, .fast').forEach(el => { el.style.animation = 'none'; }); } })();